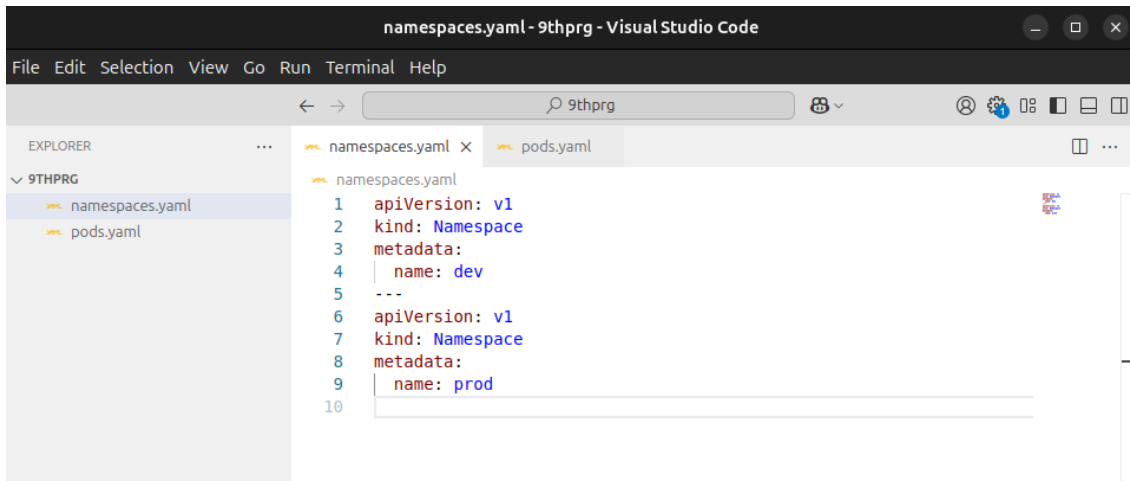


Program 9

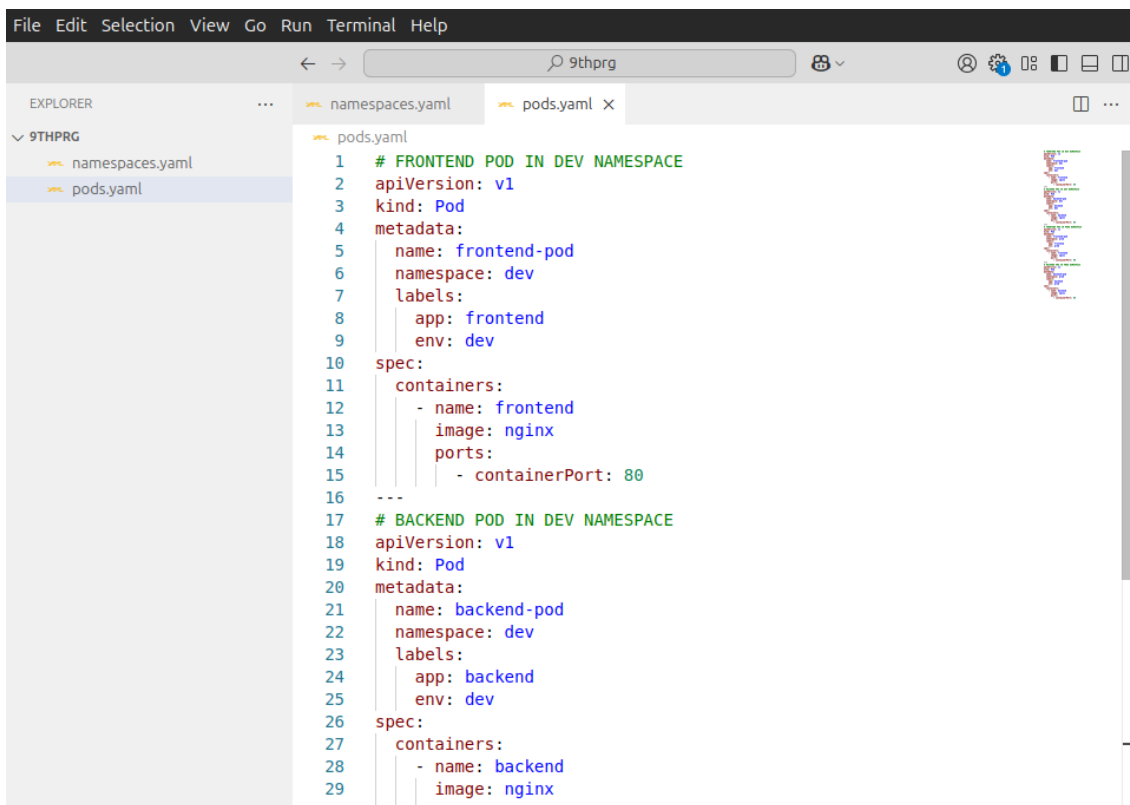
STEP 1: create a file for namespaces as namespaces.yaml



The screenshot shows the Visual Studio Code editor with the file 'namespaces.yaml' open. The Explorer sidebar on the left shows a project named '9THPRG' containing two files: 'namespaces.yaml' and 'pods.yaml'. The 'namespaces.yaml' file is selected and its content is displayed in the main editor. The content is a YAML file defining two namespaces: 'dev' and 'prod'.

```
namespaces.yaml
1  apiVersion: v1
2  kind: Namespace
3  metadata:
4    name: dev
5  ---
6  apiVersion: v1
7  kind: Namespace
8  metadata:
9    name: prod
10
```

STEP 2: Create a file for Pods as pods.yaml



The screenshot shows the Visual Studio Code editor with the file 'pods.yaml' open. The Explorer sidebar on the left shows the same project '9THPRG' with 'namespaces.yaml' and 'pods.yaml'. The 'pods.yaml' file is selected and its content is displayed in the main editor. The content is a YAML file defining two pods: 'frontend-pod' and 'backend-pod'.

```
pods.yaml
1  # FRONTEND POD IN DEV NAMESPACE
2  apiVersion: v1
3  kind: Pod
4  metadata:
5    name: frontend-pod
6    namespace: dev
7    labels:
8      app: frontend
9      env: dev
10 spec:
11   containers:
12   - name: frontend
13     image: nginx
14     ports:
15     - containerPort: 80
16   ---
17  # BACKEND POD IN DEV NAMESPACE
18  apiVersion: v1
19  kind: Pod
20  metadata:
21    name: backend-pod
22    namespace: dev
23    labels:
24      app: backend
25      env: dev
26 spec:
27   containers:
28   - name: backend
29     image: nginx
30
```

STEP 3: Apply YAML Files

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

STEP 4: Verify Pods by Namespace

STEP 5: Filter Pods Using Labels

The screenshot shows the Kubernetes dashboard interface. The top navigation bar includes the 'kubernetes' logo, a dropdown menu set to 'prod', a search bar, and a notification bell. The left sidebar contains a menu with categories: Workloads, Service, and Config and Storage. The main content area is titled 'Workload Status' and features a large green circle with the text 'Running: 2' and 'Pods' below it. Below this, a table titled 'Pods' displays the following data:

Name	Images	Labels	Node	Status
backend-pod	nginx	app: backend env: prod	minikube	Running
frontend-pod	nginx	app: frontend env: prod	minikube	Running

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           33s
frontend-pod  1/1     Running   0           33s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           39s
frontend-pod  1/1     Running   0           39s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           47s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$ kubectl get pods -l app=backend
No resources found in default namespace.
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/9thprg$
```